

Alejandro Sánchez Yalí

Senior Full Stack Engineer



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SUMMARY

Senior Full Stack Engineer with 8+ years building and maintaining scalable web applications in Python and TypeScript. Three years on a distributed team across Canada, the US and Latin America building frontend and backend and reviewing code, followed by two years as an independent consultant designing systems for clients and building three products end to end. M.Sc. in Mathematics. Strong bias toward clean architecture, testing, and decisions that survive their second year.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Backend:** Django, Django REST Framework, FastAPI, Celery, Flask, Node.js, Express, NestJS, RESTful APIs, GraphQL, microservices
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, SASS
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, Firebase
- **Cloud & DevOps:** AWS, GCP, Docker, GitHub Actions, CI/CD pipelines
- **Engineering:** Clean architecture, DDD, SOLID, DRY, TDD, Vitest, Jest, React Testing Library, code review, mentoring

EXPERIENCE

Independent AI Consultant

Freelance

Oct 2026 – Present

Remote

- Lead engineer of Aluna, an AI recruitment platform for staffing agencies (see below)

AI Specialist

Lapzo

Aug 2025 – Sep 2026

Remote, Mexico

- Made the key architectural decisions for an educational platform, designing the integration between services and the real-time communication layer
- Implemented real-time features with Next.js, NestJS and Firebase Realtime Database
- Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug, per the team's board
- Built the content-generation pipelines with Celery workers running long jobs in the background

Independent Software & AI Consultant

Freelance

Aug 2024 – Jul 2025

Remote

- Designed the database architecture for a client's workflow automation platform in an NX multi-repo, and served as code reviewer for their team
- Lead engineer of CREARIA's multi-tenant AI support agent on WhatsApp, email and Instagram, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS with Terraform, RAG over pgvector, 11 MCP tools scoped per tenant, LiteLLM fallback routing and human escalation; load-tested to 250 virtual users. Also designed the architecture of an AI coaching agent for one of CREARIA's clients
- Ran technical design sessions with clients, translating business requirements into system architectures

Full Stack Engineer

Monadical

Apr 2021 – Aug 2024

Remote, Canada/US

- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests and supports conferences of about 200 participants: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video
- Full-stack engineer on Virtue Poker, a decentralized poker platform on Ethereum supporting up to 50 concurrent tables of 8 players: React front end, Django back end and smart-contract integration
- Developed both frontend and backend, including reusable components and tests (Vitest, Jest, TDD)
- Designed ETL pipelines and custom Django ORM extensions; managed deployments on AWS and GCP with Docker and CI/CD

- Did code reviews on a distributed team in three countries

Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Designed system architecture for blockchain and analytics platforms: data models, service boundaries and smart contract patterns
- Built platforms with Python (Django), Node.js, React and Web3.js; managed infrastructure on AWS, GCP and IBM Cloud
- Mentored junior engineers, supporting their technical growth and code quality

Professor of Mathematics & Physics (part-time)

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python and the scientific stack

FEATURED PROJECTS

- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Next.js, Railway*
 - Took the product from an idea to a deployed system as architect and lead engineer: discovery, architecture, backend, frontend and deploy
 - Near-complete MVP in 4 months, covered by about 590 backend tests and CI on every pull request; infrastructure runs at \$60–80/month because cost scales with LLM usage
- **Kipux – Collaborative Finance App in Rust**  – *Rust, Axum, sqlx, PostgreSQL, HTMX, SSE, Shuttle, MIT (work in progress)*
 - Personal project to learn Rust by shipping a real app: multi-tenant projects with invites and roles, budgets, real-time updates over Server-Sent Events, and a PWA with an offline transaction queue
- **Digital Human – Interactive AI Avatar**  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Engineering case studies:** [Plixiq](#)  · [CREARIA Agent](#)  · [Aluna](#)  – architecture and trade-offs behind each system (2026)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia